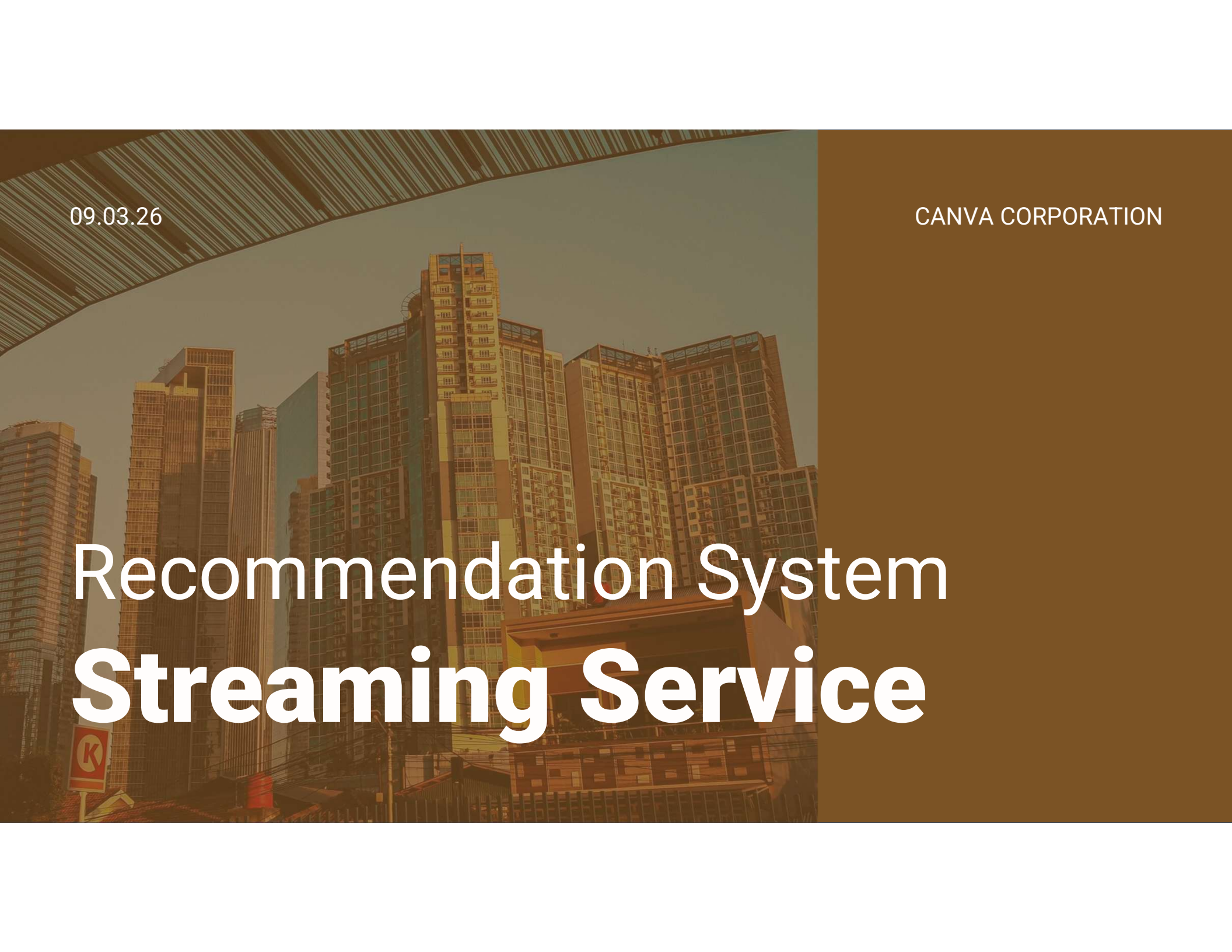


09.03.26

CANVA CORPORATION



Recommendation System **Streaming Service**



Agenda

01 Introduction

02 Data

03 Modeling

04 Recommendations and Conclusions

05 Next Steps

01

Introduction



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Recommendation System



An independent movie company is interested in entering the streaming space. The client recognizes that this space is competitive; however, the client also believes there is opportunity. As a result, the client has requested KBO Analytics to create a recommendation system.

Modeling.....

Completed the following:

- Created several prototype to serve as the recommendation system
- Identified the best performing model

02

Data



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Data Description

- Combined two csv files from MovieLens
 - *Movies.csv*
 - Columns contain the following: 1) MovieID, 2) Title, 3) Genres
 - *Ratings.csv*
 - Columns contain the following: 1) UserID, 2) MovieID, 3) (Movie) Rating

03

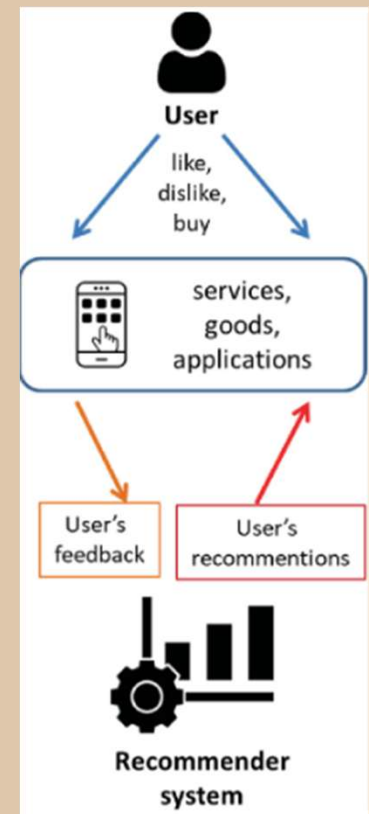
Modeling



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Single Value Decomposition

- What is Single Value Decomposition (SVD)?
 - Utilizes a combination of three algebraic matrices to make user recommendations



Model Comparisons

- Utilized Root Mean Square Error (RMSE) to measure model performance
 - Measure of the distance between the predicted values and actual values
 - the smaller the RMSE error, the better the model performance

#	Model	RMSE
1	Single Value Decomposition	0.8692
2	K-Nearest Neighbors (KNN) Basic	0.9727
3	KNN Baseline	0.8779
4	KNN with Means	0.8976

Cold-Start Problem

- Requirement to make recommendations for a new user
 - Why? → New User has not yet provided any input to the recommendation system
- Workaround → Utilize unpersonalized recommendations
 - Present the 20 most popular movies

04

Recommendations and Conclusions



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Recommendations and Conclusions

- Worked Prototype
 - However, based on dataset of $\approx 100,000$ movies

05

Next Steps

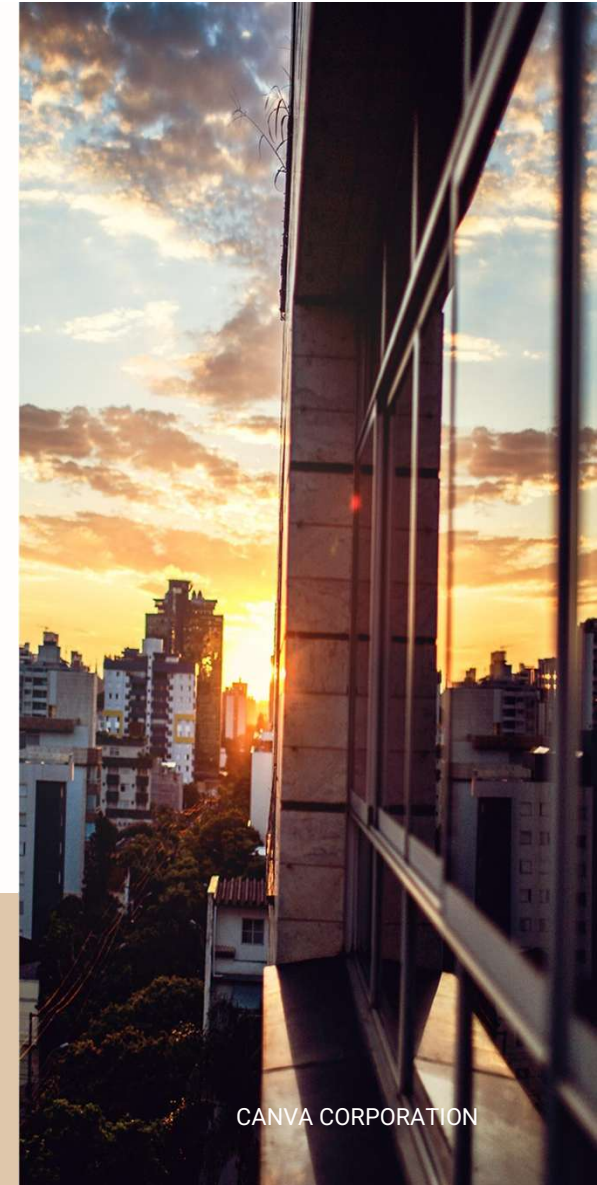


Next Steps

1 – Additional Data

2 – Acquisition

3 – Marketing and Positioning



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06

Appendix



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Cold-Start Problem | Unpersonalized Recommendations

- Top 20 Movies

Rank	Movie
1	<i>The Green Mile</i> (1999)
2	<i>Monty Python's Life of Bryan</i> (1979)
3	<i>Lightning Jack</i> (1994)
4	<i>Jurassic Park</i> (1993)
5	<i>The Flamingo Kid</i> (1984)
6	<i>Pulp Fiction</i> (1994)
7	<i>Star Wars: Episode IV – A New Hope</i> (1977)
8	<i>Fargo</i> (1996)
9	<i>Dr. Strangelove</i> (1964)
10	<i>The Godfather</i> (1972)

Rank	Movie
11	<i>12 Monkeys</i> (1995)
12	<i>The Goodbye Girl</i> (1977)
13	<i>Deathgasm</i> (2015)
14	<i>It's a Wonderful Life</i> (1946)
15	<i>Places in the Heart</i> (1984)
16	<i>Mary Poppins</i> (1964)
17	<i>The Sound of Music</i> (1965)
18	<i>Tombstone</i> (1993)
19	<i>Schindler's List</i> (1993)
20	<i>Terminator 2: Judgement Day</i> (1991)